

Milestone 2

JAVA Lab Resource Utilization Platform

Milestone Title: Utilization Monitoring & Inter-Institution Resource Sharing

Objective

The objective of Milestone 2 is to enhance the existing resource management system by implementing real-time equipment monitoring, utilization analytics, inter-institution resource sharing, external booking, and waitlist management.

Scope

- Real-time equipment utilization monitoring
- Equipment usage visualization
- Heatmap generation
- Idle equipment detection
- Inter-institution equipment sharing
- External booking system
- Demand analysis
- Waitlist management
- Booking optimization

Functional Requirements

1. Real-Time Equipment Utilization Monitoring: Track equipment status (Available, Booked, In Use, Under Maintenance), automatic updates, live dashboard.
2. Utilization Heatmap & Idle Time Detection: Daily/weekly/monthly heatmaps, idle equipment detection, color coding.

3. Inter-Institution Resource Sharing: Institution registration, shared equipment catalog, request approval/rejection workflow.

4. External Booking Management: External registration, booking, approval, cancellation and access management.

5. Utilization Rate & Demand Analysis: Utilization %, total bookings, peak usage, reports. Formula: $\text{Utilization Rate} = (\text{Hours Used} / \text{Total Available Hours}) \times 100$.

6. Waitlist Management & Booking Optimization: Automatic waitlist, queue management, notifications, alternative slot suggestions.

Frontend Modules

- Utilization Dashboard
- Equipment Status Page
- Utilization Heatmap
- Analytics Dashboard
- Shared Equipment Page
- External Booking Page
- Waitlist Page
- Booking Approval Page
- Reports Dashboard

Backend Modules

- Equipment Status Service
- Booking Service
- Utilization Service
- Analytics Service
- Sharing Service
- Waitlist Service
- Notification Service
- External Booking Service

Database Tables

- Equipment
- Booking
- Booking History

- Utilization Log
- Institution
- Shared Equipment
- Sharing Request
- External User
- Waitlist
- Notification

Technologies

- Frontend: React.js, HTML, CSS, JavaScript
- Backend: Spring Boot, Java, REST APIs
- Database: MySQL
- Tools: GitHub, VS Code, Postman, Figma

Expected Deliverables

Real-time monitoring dashboard, utilization heatmap, analytics, idle equipment detection, inter-institution sharing, external booking, waitlist management, reports.

Learning Outcomes

Develop monitoring systems, utilization analytics, sharing workflows, booking optimization, demand analysis, dashboard visualization, frontend-backend integration.

Conclusion

Milestone 2 enhances the platform with intelligent monitoring, analytics, resource sharing, and booking optimization to improve laboratory resource utilization.